Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{array}{llll} \text{minimize} & C \cdot U\hat{X}U^T & \text{maximize} & b^T y \\ \text{subject to} & A_i \cdot U\hat{X}U^T = b_i \ \ \forall i \in \{1, \dots, m\} & \text{subject to} & U^T(C - \sum_{i=1}^m y_i A_i)U \in \mathbb{S}_+^d, \\ & \hat{X} \in \mathbb{S}_+^d & & \end{array}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} q \cdot N_d &= \sum_{x,y,z \in \mathbb{F}_q} \theta(zE_d(x,y)) \\ &= q^2 + \sum_{z \in \mathbb{F}_q^\times} \theta(zb) + \sum_{y,z \in \mathbb{F}_q^\times} \theta(bz)\theta(-zy^2) + \sum_{x,z \in \mathbb{F}_q^\times} \theta(zx^d)\theta(zax)\theta(zb) \\ &+ \sum_{x,y,z \in \mathbb{F}_q^\times} \theta(x^d z)\theta(axz)\theta(bz)\theta(-zy^2) \\ &= q^2 + A + B + C + D. \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$M = \frac{1}{\sqrt{1 + \bar{\mathbf{w}} \cdot \mathbf{w} - \frac{1}{4}(\mathbf{w} \times \bar{\mathbf{w}})^2}} \begin{pmatrix} 1 - \frac{i}{2}(\mathbf{w} \times \bar{\mathbf{w}}) \cdot \sigma & -\mathbf{w} \cdot \sigma \\ beginalign*2ex]\bar{\mathbf{w}} \cdot \sigma & 1 + \frac{i}{2}(\mathbf{w} \times \bar{\mathbf{w}}) \cdot \sigma \end{pmatrix}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} U(\vec{a}, 1)\left(S, f_\alpha \otimes \rho(E^\alpha)\right) :=& e^{-i[\hat{f}_0 \cdot \hat{H}(\vec{a},\rho) + \hat{f}_1 \cdot \hat{P}(\vec{a},\rho)]} \left(S + \vec{a}, f_\alpha(\cdot - \vec{a}) \otimes \rho(E^\alpha)\right) \\ =& e^{i[a^0 \hat{H}(\rho) - a^1 \hat{P}(\rho)]} \left(S + \vec{a}, f_\alpha(\cdot - \vec{a}) \otimes \rho(E^\alpha)\right). \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$
\begin{aligned}
I_1 &= (-\infty, \omega_\xi^* - \frac{\alpha\xi^\alpha}{2(1-\alpha)}],\\
I_2 &= [\omega_\xi^* - \frac{\alpha\xi^\alpha}{2(1-\alpha)}, \omega_\xi^* + \frac{\alpha\xi^\alpha}{2(1-\alpha)}],\\
I_3 &= [\omega_\xi^* + \frac{\alpha\xi^\alpha}{2(1-\alpha)}, 0], \text{ and}\\
I_4 &= [0, \infty).
\end{aligned}
$$